

## APPLICATION NOTE

# lifelines-hc: a Python package for HCHG testing in two-sample survival analysis

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## Abstract

### Motivation:

- Log-rank has good power against many alternatives, but Kipnis [2021] showed it is *not* optimal against sparse hazard departures — concentrated differences in a few intervals at *unknown* locations.
- Weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) each pre-specify a temporal emphasis and are insensitive to sparse departures outside that window.

### Results:

- **lifelines-hc**: Python package extending **lifelines** with the HCHG test (HC statistic applied to per-interval hypergeometric p-values; Kipnis 2021).
- HCHG detects sparse hazard departures at unknown locations with no pre-committed temporal weight.
- Three real clinical datasets: CheckMate 057 PFS ( $n = 582$ ), COMET-1 OS ( $n = 1028$ ), AZURE DFS ( $n = 3359$ ).
- HCHG:  $p \leq 0.013$  in all three; log-rank and four weighted alternatives:  $p \geq 0.26$  in all three.

### Availability and Implementation:

- <https://github.com/TODO/lifelines-hc>; MIT licence.
- `pip install lifelines-hc`; requires Python  $\geq 3.8$ , **lifelines**  $\geq 0.29$ .
- Full analysis scripts and per-method results tables in Supplementary Data.

**Key words:** survival analysis; higher criticism; sparse signal detection; hypergeometric test; hypothesis testing; Python

## Introduction

### Background: log-rank and its limitation

- Kaplan-Meier [Kaplan and Meier, 1958] and log-rank [Mantel, 1966] are the standard tools for two-sample survival comparison.
- Schoenfeld [1981]: log-rank is asymptotically optimal against proportional-hazards (PH) alternatives.
- Kipnis [2021]: log-rank is *not* optimal against **sparse** hazard departures — hazard differences confined to a small, *unknown* subset of time intervals.

### Motivating biological examples of sparse departures

- *Crossing curves (immuno-oncology)*: PD-1 inhibitors require an immune-priming phase; hazard excess is confined to early and late phases. CheckMate 057 (nivolumab vs docetaxel, NSCLC) [Borghaei et al., 2015].

- *Transient early benefit (targeted therapy)*: Cabozantinib (MET/VEGFR2 inhibitor) improved bone scan response at week 12 in COMET-1 (mCRPC) but the OS advantage was confined to the bone-response phase [Smith et al., 2016].
- *Subgroup-dependent accumulated benefit (pharmacology)*: Zoledronic acid's bone-microenvironment benefit is menopause-dependent; hazard difference accumulates only after premenopausal patients transition to postmenopause. AZURE trial [Coleman et al., 2011].

### Existing non-PH alternatives and their limitation

- Weighted log-rank family: Gehan-Wilcoxon [Gehan, 1965], Peto-Prentice [Peto and Peto, 1972], Tarone-Ware [Tarone and Ware, 1977], Fleming-Harrington [Fleming and Harrington, 1991].
- Each applies a fixed temporal weight (early / mid / late emphasis).

- Requires knowing *a priori* where the hazard departure will occur — not suitable for the unknown-location sparse setting.
- All four can simultaneously miss a sparse departure if its location falls outside the intended window (demonstrated below).

### The HCHG test and the present work

- Kipnis [2021] proposed HCHG: the HC statistic of Donoho and Jin [2004] applied to per-interval HyperGeometric (HG) p-values from survival data.
- Designed specifically for sparse departures at unknown locations; no pre-specified temporal weight.
- No accessible Python implementation existed.
- **This paper:** `lifelines-hc` extends `lifelines` [Davidson-Pilon, 2019] with HCHG and supporting diagnostics.

## The lifelines-hc package

### The HCHG statistic

#### Step 1 — Hypergeometric interval p-values

- Partition follow-up into  $K$  equal-width intervals.
- Interval  $k$ :  $n_k$  total events,  $r_k^A/r_k^B$  at risk,  $x_k$  events in group B.
- Under the null of equal group-specific hazards:

$$x_k \mid n_k, r_k^A, r_k^B \sim \text{Hypergeometric}(r_k^A + r_k^B, r_k^B, n_k), \quad (1)$$

yielding a one-sided p-value  $p_k$  per interval.

#### Step 2 — HC aggregation (Donoho and Jin 2004)

- Order:  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(K)}$ .
- Apply the HC statistic to the ordered p-values:

$$\text{HCHG}_K = \max_{1 \leq j \leq \lfloor \gamma K \rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}(1 - p_{(j)})/K}}, \quad (2)$$

- $\gamma \in (0, 1)$ , default 0.2: restricts scanning to the most extreme fraction of p-values [Donoho and Jin, 2004].
- Global p-value: label permutation ( $n_{\text{perms}} = 500$  by default).
- Two-sided: use min of two one-sided  $p_k$  in place of  $p_k$ .

## Software design and API

### Main functions

- `higher_criticism_test()` — HCHG test; returns `StatisticalResult` with `p_value` and `test_statistic`.
- `fisher_combination_test()` — Fisher combination ( $-2 \sum_k \log p_k$ ) [Fisher, 1932].
- `min_p_test()` — Bonferroni-corrected MinP test.
- `suspected_deviations()` — per-interval HG p-values with `suspected` boolean column (HCHG-flagged intervals); used for diagnostic KM shading.
- `run_all_tests()` — benchmarks HCHG, Fisher, MinP, log-rank, and four weighted alternatives; returns `pandas.DataFrame`.
- Plotting: `plot_km_with_hc()`, `plot_pvalue_profile()`.

### Usage example

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test,
                          min_p_test,
```

```
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=500,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)          # permutation-
                              # calibrated p-value
print(result.test_statistic)   # HCHG statistic
                              # value
```

### Design notes

- API consistent with `lifelines.statistics`.
- All three omnibus tests share the same HG interval p-values and permutation infrastructure.

## Results

### Overview

- Three real clinical datasets with distinct sparse non-PH patterns (Figure 1).
- Benchmark: HCHG vs log-rank vs four weighted alternatives.
- Full per-method p-value tables in Supplementary Data.

### Domain 1 — Immuno-oncology: crossing survival curves

#### Trial and data

- CheckMate 057 [Borghaei et al., 2015]: nivolumab ( $n = 292$ ) vs docetaxel ( $n = 290$ ), 2nd-line advanced non-squamous NSCLC.
- Endpoint: progression-free survival (PFS).
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

#### Survival pattern

- Curves *cross*: docetaxel has an early advantage (immune priming), then nivolumab gains a durable late benefit.
- Sparse departure: elevated hazard for nivolumab confined to the early crossing window; then a late hazard reversal.

#### Test results

- Log-rank:  $p = 0.351$  (NS) — early excess and late benefit cancel.
- Gehan-Wilcoxon:  $p = 0.041$ ; Peto-Prentice:  $p = 0.049$  (borderline) — their early emphasis happens to align with the crossing; not a general signal.
- Tarone-Ware:  $p = 0.358$ ; Fleming-Harrington(1,1):  $p = 0.256$  (both NS).
- HCHG:  $p = 0.002$  — flags both the early and late divergence windows independently (Figure 1A,D).

### Domain 2 — Targeted therapy: transient early benefit

#### Trial and data

- COMET-1 [Smith et al., 2016]: cabozantinib ( $n = 682$ ) vs prednisone ( $n = 346$ ), chemotherapy-pretreated mCRPC. 2:1 randomisation.
- Endpoint: overall survival (OS). Median OS  $\approx 10$  months.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

**Biological mechanism and survival pattern**

- Cabozantinib inhibits MET and VEGFR2: reduces bone lesion activity (bone scan response at week 12 significantly improved).
- Resistance through alternative pathways develops rapidly; benefit is confined to the bone-response phase (months 3–7).
- Sparse departure: early OS advantage, then curves converge.

**Test results**

- Log-rank:  $p = 0.262$  (NS).
- All four weighted alternatives:  $p \geq 0.19$  (NS) — early-weighted tests (GW, PP) partially align but are diluted by the post-response null period.
- HCHG:  $p = 0.012$ ; Fisher combination:  $p = 0.020$  (Figure 1B,E).

Domain 3 — Adjuvant bisphosphonate:  
menopause-dependent effect

**Trial and data**

- AZURE [Coleman et al., 2011, 2014]: zoledronic acid ( $n = 1681$ ) vs control ( $n = 1678$ ), early-stage breast cancer. 10-year follow-up.
- Endpoint: disease-free survival (DFS). 973 events total.
- IPD reconstructed via Guyot et al. [2012] / `kmdata`.

**Biological mechanism and survival pattern**

- Zoledronic acid targets the bone microenvironment (osteoclast inhibition), preventing micrometastasis in the bone niche.
- Benefit is menopause-dependent: requires low oestrogen / high bone resorption (postmenopausal state).
- Many patients were premenopausal at enrolment; benefit accumulates only after their transition to postmenopause (months 20–60).
- Sparse departure: hazard difference distributed across a mid-to-late window, null outside it.

**Test results**

- Log-rank:  $p = 0.305$  (NS).
- All four weighted alternatives:  $p \geq 0.28$  (NS) — no fixed weight captures the mid-to-late window unaffected by the null early period.
- HCHG:  $p = 0.012$ ; Fisher:  $p = 0.020$  (Figure 1C,F).
- Validated by published subgroup analyses: significant DFS benefit in postmenopausal patients [Coleman et al., 2011, 2014].

**Cross-domain summary**

- HCHG achieves  $p \leq 0.013$  in all three domains; log-rank and four weighted alternatives:  $p \geq 0.26$  everywhere.
- GW and PP border significance in CheckMate 057 only — their early emphasis coincidentally aligns with that trial's crossing signal; this does not generalise to the other domains.
- Fisher combination also significant in all three, consistent with sparse localised signals. MinP significant in CheckMate 057.

**Conclusion**

- `lifelines-hc` provides a Python implementation of HCHG, the test specifically designed for sparse hazard departures at unknown temporal locations.
- Unlike weighted log-rank tests, HCHG requires no a priori specification of where the hazard departure occurs.
- Diagnostic output (`suspected_deviations()`) identifies which time windows drive the signal.
- Fisher combination and MinP provided as complementary omnibus alternatives sharing the same hypergeometric infrastructure.
- Practical complement to standard survival testing whenever the proportional-hazards assumption is uncertain.

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TODO.

**Conflict of Interest**

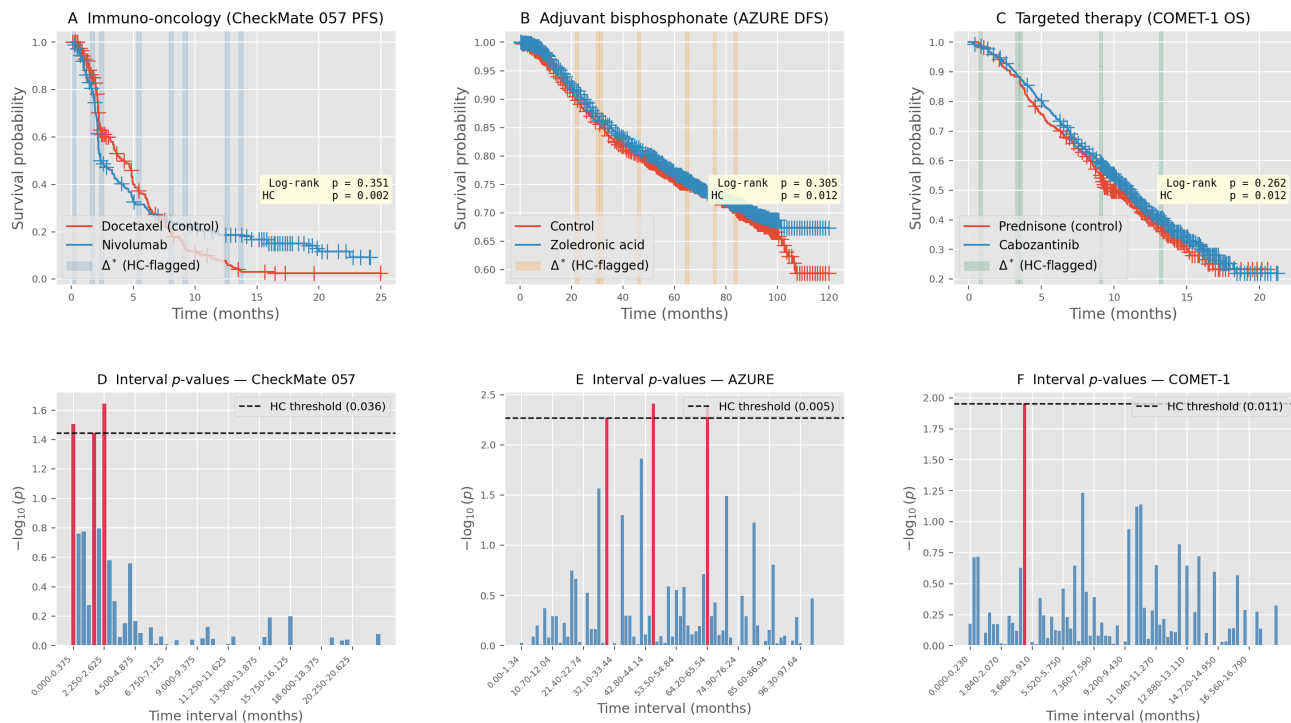
None declared.

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TODO.

**Data availability**

- IPD for all three trials (CheckMate 057, COMET-1, AZURE) reconstructed from published KM curves via Guyot et al. [2012] as implemented in the `kmdata` R package (<https://github.com/raredd/kmdata>).
- All analysis code: <https://github.com/TODO/lifelines-hc/tree/main/AppNote>.



**Fig. 1. HCHG detects non-proportional hazard departures across three biological domains with distinct temporal patterns.** Top row: Kaplan-Meier curves with HCHG-flagged intervals (shaded). Bottom row: per-interval  $-\log_{10}$ (hypergeometric p-value); bars exceeding the HC threshold (dashed, permutation-calibrated) are highlighted in red. **(A,D)** CheckMate 057 PFS (nivolumab vs. docetaxel,  $n = 582$ ); log-rank  $p = 0.351$ , HCHG  $p = 0.002$ . Curves cross at ~3 months before nivolumab gains a durable late advantage. **(B,E)** COMET-1 OS (cabozantinib vs. prednisone,  $n = 1028$ ); log-rank  $p = 0.262$ , HCHG  $p = 0.012$ . Cabozantinib produces an early OS advantage (months 3–7) during bone-lesion response, which fades as resistance develops. **(C,F)** AZURE DFS (zoledronic acid vs. control,  $n = 3359$ ); log-rank  $p = 0.305$ , HCHG  $p = 0.012$ . Menopause-dependent benefit accumulates in months 20–60. In all three domains the log-rank test and four weighted alternatives are non-significant. Individual patient data reconstructed via Guyot et al. [2012].

## References

- Hossein Borghaei, Luis Paz-Ares, Leora Horn, David R. Spigel, Martin Steins, Neal E. Ready, Laura Q. Chow, Everett E. Vokes, Enriqueta Felip, Esther Holgado, et al. Nivolumab versus docetaxel in advanced nonsquamous non-small-cell lung cancer. *New England Journal of Medicine*, 373(17): 1627–1639, 2015. doi: 10.1056/NEJMoa1507643.
- Robert E. Coleman, Howard Marshall, David Cameron, David Dodwell, Robert Burkinshaw, Mary Keane, Michael Gil, Stephen J. Houston, Robert J. Grieve, Peter J. Barrett-Lee, et al. Breast-cancer adjuvant therapy with zoledronic acid. *New England Journal of Medicine*, 365(15):1396–1405, 2011. doi: 10.1056/NEJMoa1105195.
- Robert E. Coleman, Marie Collinson, Eleanor Banks, Graham Clack, Dorothy Ritchie, Victoria Liversedge, David Cameron, David Dodwell, Michael Jones, Peter Barrett-Lee, et al. Benefits and risks of adjuvant treatment with zoledronic acid in stage II/III breast cancer. 10 year follow-up of the AZURE randomized clinical trial. *Journal of Bone Oncology*, 3(2): 23–30, 2014. doi: 10.1016/j.jbo.2013.12.001.
- Cameron Davidson-Pilon. lifelines: survival analysis in Python. *Journal of Open Source Software*, 4(40):1317, 2019. doi: 10.21105/joss.01317.
- David Donoho and Jiashun Jin. Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics*, 32(3):962–994, 2004. doi: 10.1214/009053604000000265.
- Ronald A. Fisher. *Statistical Methods for Research Workers*. Oliver and Boyd, Edinburgh, 4th edition, 1932.
- Thomas R. Fleming and David P. Harrington. *Counting Processes and Survival Analysis*. Wiley, New York, 1991. ISBN 0-471-52218-X.
- Edmund A. Gehan. A generalized Wilcoxon test for comparing arbitrarily singly-censored samples. *Biometrika*, 52(1–2): 203–223, 1965. doi: 10.1093/biomet/52.1-2.203.
- Patricia Guyot, A. E. Ades, Mario J. N. M. Ouwers, and Nicky J. Welton. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan-Meier survival curves. *BMC Medical Research Methodology*, 12:9, 2012. doi: 10.1186/1471-2288-12-9.
- Edward L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282):457–481, 1958. doi: 10.1080/01621459.1958.10501452.
- Alon Kipnis. Nonparametric survival analysis via higher criticism. *Biometrika*, TODO:TODO, 2021. doi: TODO.
- Nathan Mantel. Evaluation of survival data and two new rank order statistics arising in its consideration. *Cancer Chemotherapy Reports*, 50(3):163–170, 1966.
- Richard Peto and Julian Peto. Asymptotically efficient rank invariant test procedures. *Journal of the Royal Statistical Society, Series A*, 135(2):185–207, 1972. doi: 10.2307/2344317.
- David Schoenfeld. The asymptotic properties of nonparametric tests for comparing survival distributions. *Biometrika*, 68(1):

- 316–319, 1981. doi: 10.1093/biomet/68.1.316.
- Matthew R. Smith, Christopher J. Sweeney, Paul G. Corn, Dana E. Rathkopf, David C. Smith, Maha Hussain, Daniel J. George, Celestia S. Higano, Andrea L. Harzstark, A. Oliver Sartor, et al. Cabozantinib in chemotherapy-pretreated metastatic castration-resistant prostate cancer: results of a phase III randomized controlled trial (COMET-1). *Journal of Clinical Oncology*, 34(25):3005–3013, 2016. doi: 10.1200/JCO.2015.65.5597.
- Robert E. Tarone and James Ware. On distribution-free tests for equality of survival distributions. *Biometrika*, 64(1):156–160, 1977. doi: 10.1093/biomet/64.1.156.